

# MLib Build Automation

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MHI

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## Problem: Library Building & Merging

- Merging the Library is Manual & Time Consuming
- Library Builds are Slow

## Tactical Solution: Automation of ...

- SVN Update
- Library Build Process
- Running of Tests

## 1 Library Build Scripts

- SVN Update
- Build Library
- Run Tests

## 2 Combining Build Scripts

- Update then BuildAll then TestAll

## Automation using Build Scripts

- Build scripts or batch files have been written to quickly update, build the library and to run the library tests. Scripts are available in the new %MLIB%/Utilities/BatchFiles folder.

## Tactical Scripts Available

- ① SVN Update
- ② Build Library (All Configs in Parallel)
- ③ Run Tests (All Configs in Parallel)

## Benefits

- ① Faster Builds x3
- ② Supports Build Scheduling, using windows Task Scheduler
- ③ Scripts are compatible with continuous integration tools

## The Plan

- The tactical build scripts will be extended further and have been set-up to be compatible with continuous integration tools. We are considering using Jenkins, which is free-ware.

## Next Tasks

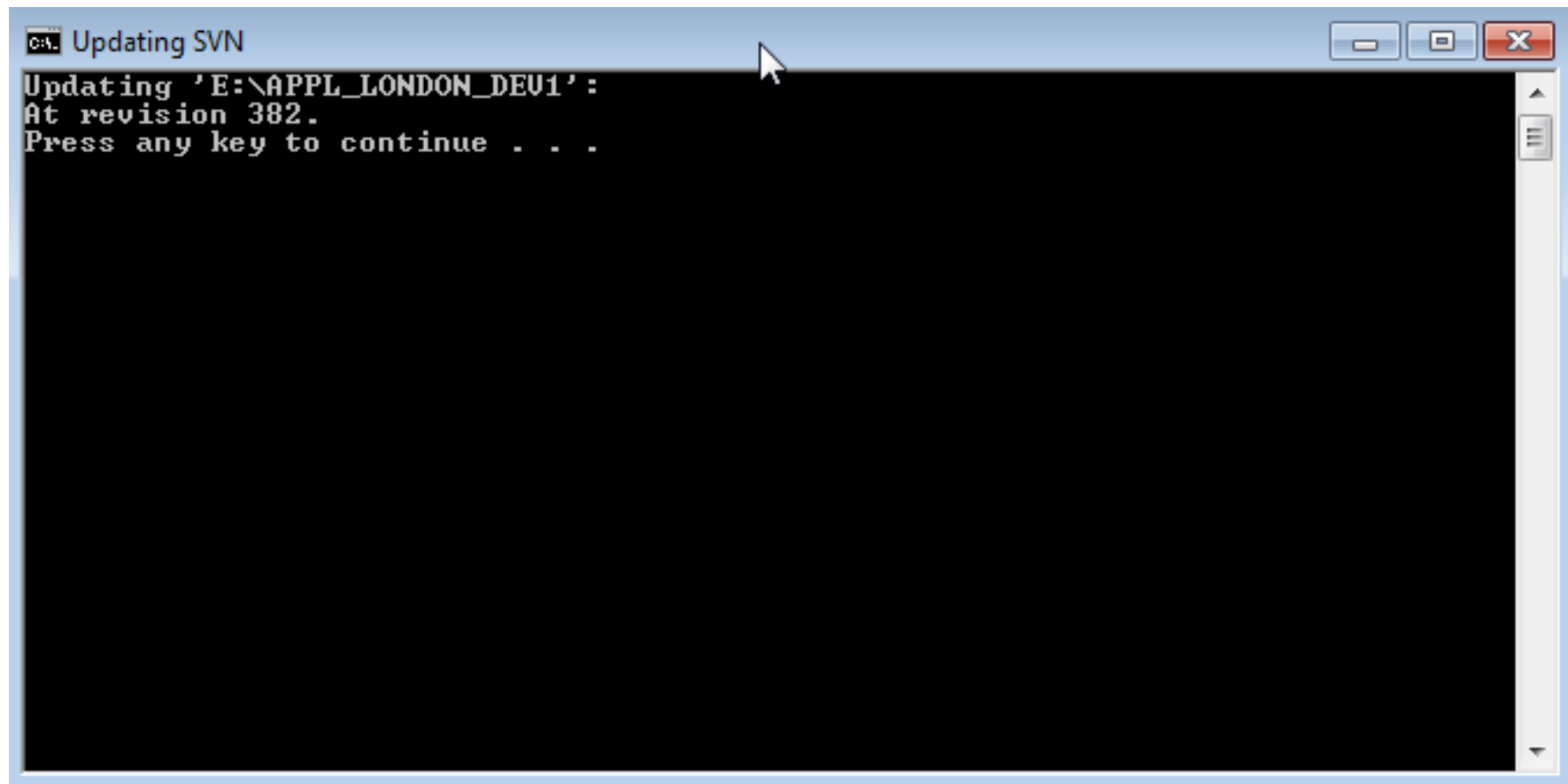
- ① Create Build & Test Log Files
- ② Revert Updates & / or Merges if Build or Tests Fail
- ③ Merge Branches to UAT
- ④ Merge UAT to Trunk

## Helpful Tools

- Use of Solid State Drives (Faster Build x2)
- Use of Distributed Build Software e.g. Incredibuild (Faster Build x5)
- Use of Continuous Integration Tools e.g. Jenkins, Bamboo, Team City ...

## Batch File: Update.bat

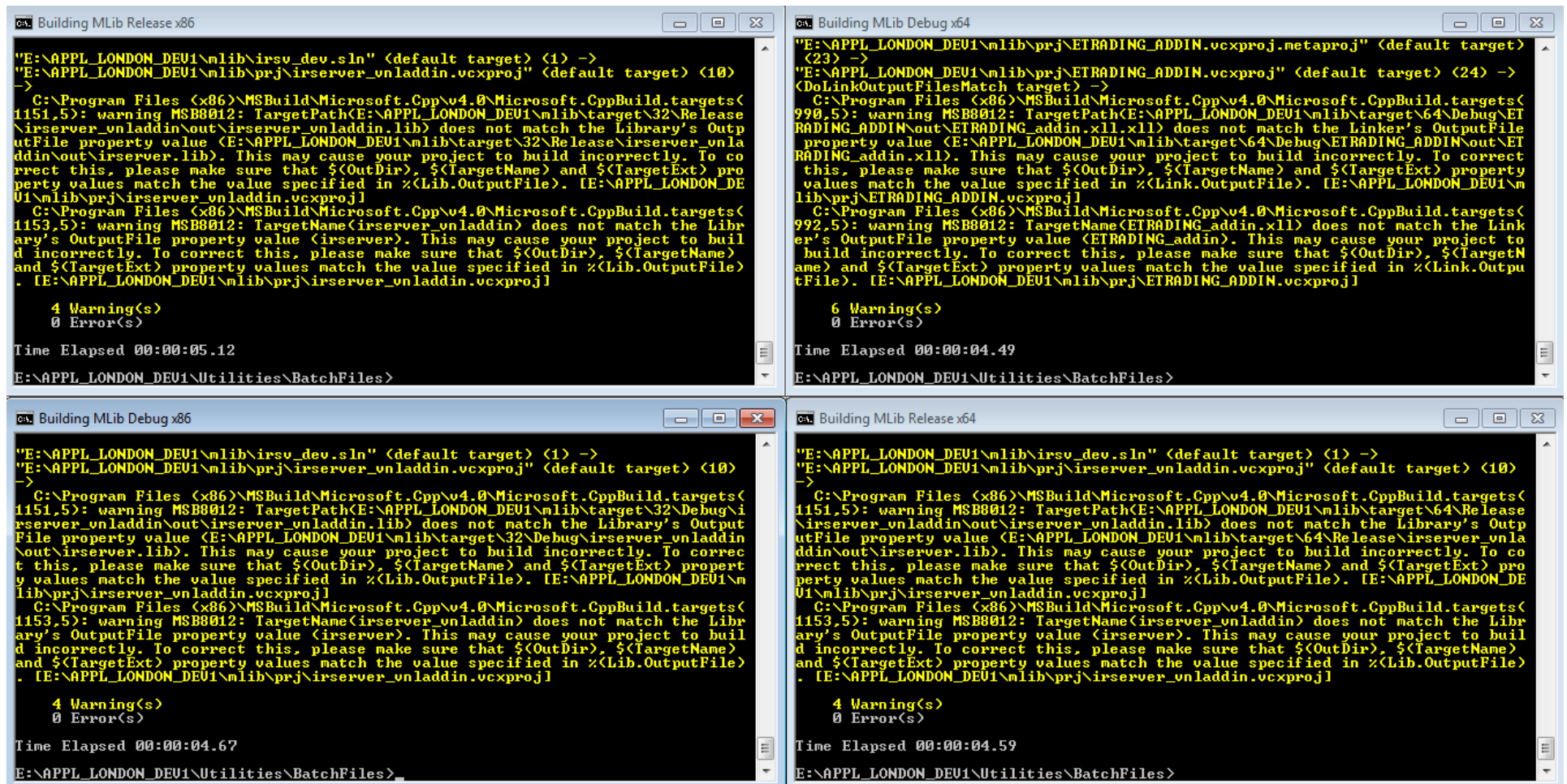
- This batch file will run SVN update your working copy of the library. Just double-click or open the file to run.



```
Updating SVN
Updating 'E:\APPL_LONDON_DEV1':
At revision 382.
Press any key to continue . . .
```

## Batch File: BuildAll.bat

- This batch file will build all library configurations and platforms in parallel. Just double-click or open the file to run.



```
"E:\APPL_LONDON_DEU1\mli\rsu_dev.sln" (default target) (1) ->
"E:\APPL_LONDON_DEU1\mli\prj\irserver_unladdin.vcxproj" (default target) (10) ->
C:\Program Files (x86)\MSBuild\Microsoft.Cpp\v4.0\Microsoft.CppBuild.targets(1151,5): warning MSB8012: TargetPath(E:\APPL_LONDON_DEU1\mli\target\32\Release\irserver_unladdin\out\irserver_unladdin.lib) does not match the Library's OutputFile property value (E:\APPL_LONDON_DEU1\mli\target\32\Release\irserver_unladdin\out\irserver.lib). This may cause your project to build incorrectly. To correct this, please make sure that $(OutDir), $(TargetName) and $(TargetExt) property values match the value specified in $(Lib.OutputFile). [E:\APPL_LONDON_DEU1\mli\prj\irserver_unladdin.vcxproj]
C:\Program Files (x86)\MSBuild\Microsoft.Cpp\v4.0\Microsoft.CppBuild.targets(1153,5): warning MSB8012: TargetName(irserver_unladdin) does not match the Library's OutputFile property value (irserver). This may cause your project to build incorrectly. To correct this, please make sure that $(OutDir), $(TargetName) and $(TargetExt) property values match the value specified in $(Lib.OutputFile). [E:\APPL_LONDON_DEU1\mli\prj\irserver_unladdin.vcxproj]
4 Warning(s)
0 Error(s)
Time Elapsed 00:00:05.12
E:\APPL_LONDON_DEU1\Utilities\BatchFiles>

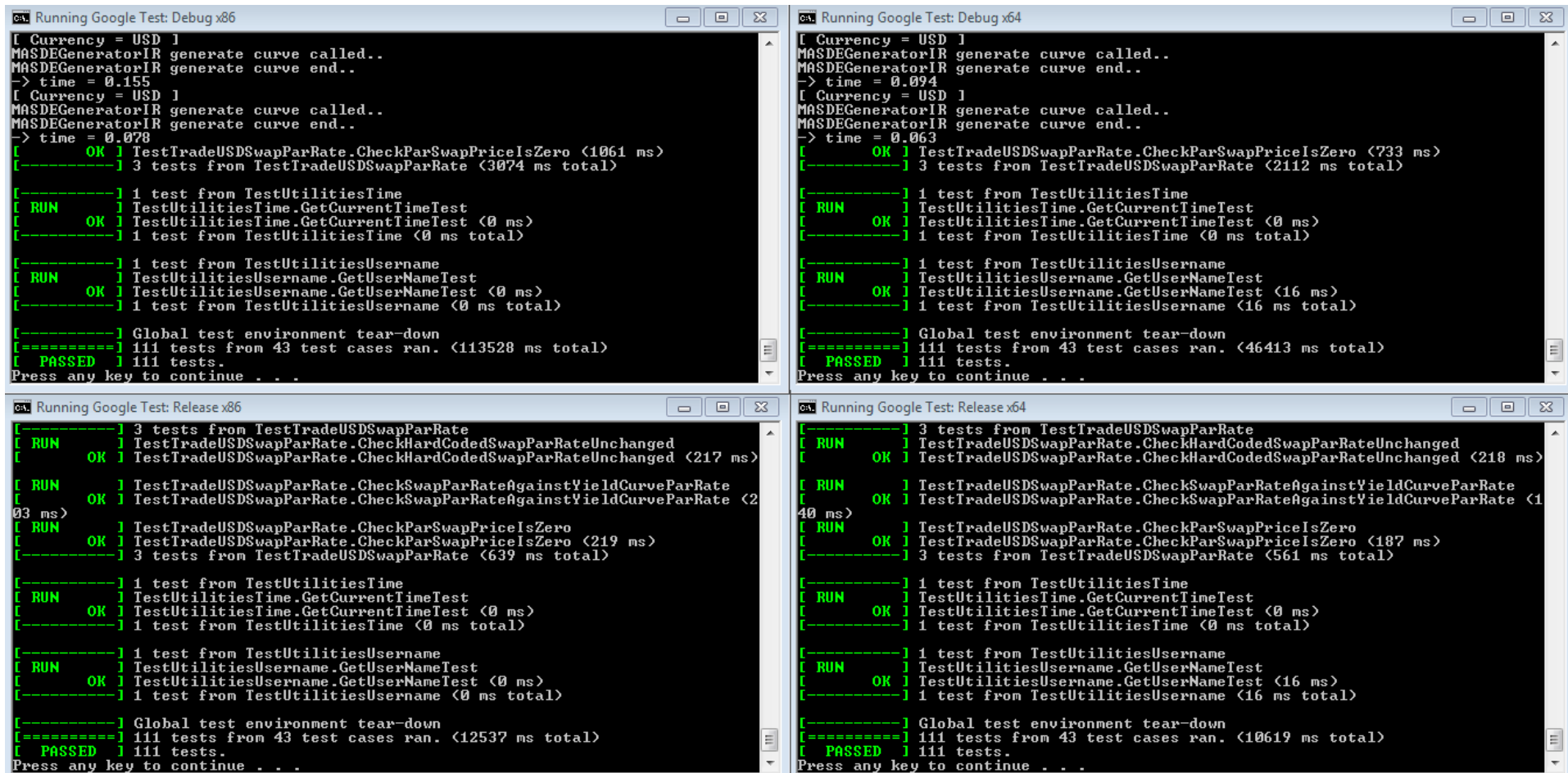
"E:\APPL_LONDON_DEU1\mli\rsu_dev.sln" (default target) (1) ->
"E:\APPL_LONDON_DEU1\mli\prj\irserver_unladdin.vcxproj" (default target) (10) ->
C:\Program Files (x86)\MSBuild\Microsoft.Cpp\v4.0\Microsoft.CppBuild.targets(1151,5): warning MSB8012: TargetPath(E:\APPL_LONDON_DEU1\mli\target\64\Debug\ETRADING_ADDIN\out\ETRADING_addin.xll) does not match the Linker's OutputFile property value (E:\APPL_LONDON_DEU1\mli\target\64\Debug\ETRADING_ADDIN\out\ETRADING_addin.xll). This may cause your project to build incorrectly. To correct this, please make sure that $(OutDir), $(TargetName) and $(TargetExt) property values match the value specified in $(Link.OutputFile). [E:\APPL_LONDON_DEU1\mli\prj\ETRADING_ADDIN.vcxproj]
C:\Program Files (x86)\MSBuild\Microsoft.Cpp\v4.0\Microsoft.CppBuild.targets(992,5): warning MSB8012: TargetName(ETRADING_addin.xll) does not match the Linker's OutputFile property value (ETRADING_addin). This may cause your project to build incorrectly. To correct this, please make sure that $(OutDir), $(TargetName) and $(TargetExt) property values match the value specified in $(Link.OutputFile). [E:\APPL_LONDON_DEU1\mli\prj\ETRADING_ADDIN.vcxproj]
6 Warning(s)
0 Error(s)
Time Elapsed 00:00:04.49
E:\APPL_LONDON_DEU1\Utilities\BatchFiles>

"E:\APPL_LONDON_DEU1\mli\rsu_dev.sln" (default target) (1) ->
"E:\APPL_LONDON_DEU1\mli\prj\irserver_unladdin.vcxproj" (default target) (10) ->
C:\Program Files (x86)\MSBuild\Microsoft.Cpp\v4.0\Microsoft.CppBuild.targets(1151,5): warning MSB8012: TargetPath(E:\APPL_LONDON_DEU1\mli\target\32\Debug\irserver_unladdin\out\irserver_unladdin.lib) does not match the Library's OutputFile property value (E:\APPL_LONDON_DEU1\mli\target\32\Debug\irserver_unladdin\out\irserver.lib). This may cause your project to build incorrectly. To correct this, please make sure that $(OutDir), $(TargetName) and $(TargetExt) property values match the value specified in $(Lib.OutputFile). [E:\APPL_LONDON_DEU1\mli\prj\irserver_unladdin.vcxproj]
C:\Program Files (x86)\MSBuild\Microsoft.Cpp\v4.0\Microsoft.CppBuild.targets(1153,5): warning MSB8012: TargetName(irserver_unladdin) does not match the Library's OutputFile property value (irserver). This may cause your project to build incorrectly. To correct this, please make sure that $(OutDir), $(TargetName) and $(TargetExt) property values match the value specified in $(Lib.OutputFile). [E:\APPL_LONDON_DEU1\mli\prj\irserver_unladdin.vcxproj]
4 Warning(s)
0 Error(s)
Time Elapsed 00:00:04.67
E:\APPL_LONDON_DEU1\Utilities\BatchFiles>

"E:\APPL_LONDON_DEU1\mli\rsu_dev.sln" (default target) (1) ->
"E:\APPL_LONDON_DEU1\mli\prj\irserver_unladdin.vcxproj" (default target) (10) ->
C:\Program Files (x86)\MSBuild\Microsoft.Cpp\v4.0\Microsoft.CppBuild.targets(1151,5): warning MSB8012: TargetPath(E:\APPL_LONDON_DEU1\mli\target\64\Release\irserver_unladdin\out\irserver_unladdin.lib) does not match the Library's OutputFile property value (E:\APPL_LONDON_DEU1\mli\target\64\Release\irserver_unladdin\out\irserver.lib). This may cause your project to build incorrectly. To correct this, please make sure that $(OutDir), $(TargetName) and $(TargetExt) property values match the value specified in $(Lib.OutputFile). [E:\APPL_LONDON_DEU1\mli\prj\irserver_unladdin.vcxproj]
C:\Program Files (x86)\MSBuild\Microsoft.Cpp\v4.0\Microsoft.CppBuild.targets(1153,5): warning MSB8012: TargetName(irserver_unladdin) does not match the Library's OutputFile property value (irserver). This may cause your project to build incorrectly. To correct this, please make sure that $(OutDir), $(TargetName) and $(TargetExt) property values match the value specified in $(Lib.OutputFile). [E:\APPL_LONDON_DEU1\mli\prj\irserver_unladdin.vcxproj]
4 Warning(s)
0 Error(s)
Time Elapsed 00:00:04.59
E:\APPL_LONDON_DEU1\Utilities\BatchFiles>
```

## Build Script: TestAll.bat

- This batch file will run all library tests for all configurations and platforms in parallel. Just double-click or open the file to run.



```
Running Google Test: Debug x86
[ Currency = USD ]
MASDEGeneratorIR generate curve called..
MASDEGeneratorIR generate curve end..
-> time = 0.155
[ Currency = USD ]
MASDEGeneratorIR generate curve called..
MASDEGeneratorIR generate curve end..
-> time = 0.078
[ OK ] TestTradeUSDSwapParRate.CheckParSwapPriceIsZero (1061 ms)
[ OK ] 3 tests from TestTradeUSDSwapParRate (3074 ms total)

[ RUN ] 1 test from TestUtilitiesTime
[ OK ] TestUtilitiesTime.GetCurrentTimeTest
[ OK ] TestUtilitiesTime.GetCurrentTimeTest (0 ms)
[ OK ] 1 test from TestUtilitiesTime (0 ms total)

[ RUN ] 1 test from TestUtilitiesUsername
[ OK ] TestUtilitiesUsername.GetUserNameTest
[ OK ] TestUtilitiesUsername.GetUserNameTest (0 ms)
[ OK ] 1 test from TestUtilitiesUsername (0 ms total)

[ Global test environment tear-down ]
[ OK ] 111 tests from 43 test cases ran. (113528 ms total)
[ PASSED ] 111 tests.
Press any key to continue . . .

Running Google Test: Release x86
[ RUN ] 3 tests from TestTradeUSDSwapParRate
[ OK ] TestTradeUSDSwapParRate.CheckHardCodedSwapParRateUnchanged (217 ms)
[ OK ] TestTradeUSDSwapParRate.CheckHardCodedSwapParRateUnchanged (217 ms)
[ RUN ] TestTradeUSDSwapParRate.CheckSwapParRateAgainstYieldCurveParRate (203 ms)
[ OK ] TestTradeUSDSwapParRate.CheckSwapParRateAgainstYieldCurveParRate (203 ms)
[ RUN ] TestTradeUSDSwapParRate.CheckParSwapPriceIsZero (219 ms)
[ OK ] TestTradeUSDSwapParRate.CheckParSwapPriceIsZero (219 ms)
[ OK ] 3 tests from TestTradeUSDSwapParRate (639 ms total)

[ RUN ] 1 test from TestUtilitiesTime
[ OK ] TestUtilitiesTime.GetCurrentTimeTest
[ OK ] TestUtilitiesTime.GetCurrentTimeTest (0 ms)
[ OK ] 1 test from TestUtilitiesTime (0 ms total)

[ RUN ] 1 test from TestUtilitiesUsername
[ OK ] TestUtilitiesUsername.GetUserNameTest
[ OK ] TestUtilitiesUsername.GetUserNameTest (0 ms)
[ OK ] 1 test from TestUtilitiesUsername (0 ms total)

[ Global test environment tear-down ]
[ OK ] 111 tests from 43 test cases ran. (12537 ms total)
[ PASSED ] 111 tests.
Press any key to continue . . .

Running Google Test: Debug x64
[ Currency = USD ]
MASDEGeneratorIR generate curve called..
MASDEGeneratorIR generate curve end..
-> time = 0.094
[ Currency = USD ]
MASDEGeneratorIR generate curve called..
MASDEGeneratorIR generate curve end..
-> time = 0.063
[ OK ] TestTradeUSDSwapParRate.CheckParSwapPriceIsZero (733 ms)
[ OK ] 3 tests from TestTradeUSDSwapParRate (2112 ms total)

[ RUN ] 1 test from TestUtilitiesTime
[ OK ] TestUtilitiesTime.GetCurrentTimeTest
[ OK ] TestUtilitiesTime.GetCurrentTimeTest (0 ms)
[ OK ] 1 test from TestUtilitiesTime (0 ms total)

[ RUN ] 1 test from TestUtilitiesUsername
[ OK ] TestUtilitiesUsername.GetUserNameTest
[ OK ] TestUtilitiesUsername.GetUserNameTest (16 ms)
[ OK ] 1 test from TestUtilitiesUsername (16 ms total)

[ Global test environment tear-down ]
[ OK ] 111 tests from 43 test cases ran. (46413 ms total)
[ PASSED ] 111 tests.
Press any key to continue . . .

Running Google Test: Release x64
[ RUN ] 3 tests from TestTradeUSDSwapParRate
[ OK ] TestTradeUSDSwapParRate.CheckHardCodedSwapParRateUnchanged (218 ms)
[ OK ] TestTradeUSDSwapParRate.CheckHardCodedSwapParRateUnchanged (218 ms)
[ RUN ] TestTradeUSDSwapParRate.CheckSwapParRateAgainstYieldCurveParRate (140 ms)
[ OK ] TestTradeUSDSwapParRate.CheckSwapParRateAgainstYieldCurveParRate (140 ms)
[ RUN ] TestTradeUSDSwapParRate.CheckParSwapPriceIsZero (187 ms)
[ OK ] TestTradeUSDSwapParRate.CheckParSwapPriceIsZero (187 ms)
[ OK ] 3 tests from TestTradeUSDSwapParRate (561 ms total)

[ RUN ] 1 test from TestUtilitiesTime
[ OK ] TestUtilitiesTime.GetCurrentTimeTest
[ OK ] TestUtilitiesTime.GetCurrentTimeTest (0 ms)
[ OK ] 1 test from TestUtilitiesTime (0 ms total)

[ RUN ] 1 test from TestUtilitiesUsername
[ OK ] TestUtilitiesUsername.GetUserNameTest
[ OK ] TestUtilitiesUsername.GetUserNameTest (16 ms)
[ OK ] 1 test from TestUtilitiesUsername (16 ms total)

[ Global test environment tear-down ]
[ OK ] 111 tests from 43 test cases ran. (10619 ms total)
[ PASSED ] 111 tests.
Press any key to continue . . .
```

## Combined Batch Files

- Batch files can be combined and scheduled to perform any combination of tasks in any order

## Available Script Combinations

- UpdateThenBuildAll.bat
- UpdateThenBuildAllThenTestAll.bat

# The End